TEXTBOOK · FALL 2026

Linear Algebra I
Weeks 1–4

All questions and all formulas from every course file and the assigned textbook sections — vectors, linear combinations and span; dot and cross products; matrices, matrix algebra, determinants; rank and the inverse matrix — each problem with a complete step-by-step solution.
Compiled for independent study and Test 1 preparation

Tashkent Technological University · Fall 2026
Sources: course notes, lectures, ISA, TA sessions;
G. Strang, Introduction to Linear Algebra (4e) and
Linear Algebra and its Applications (4e)

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1 Course Map and How to Use This Document

This document is a complete standalone guide for the first four weeks of the **Linear Algebra I** course in the first semester (Fall 2026). It was manually compiled from each relevant source in the course-files/ folder of the course repository: weekly notes (Notes), lecture presentations, exercise sheets for practical sessions, independent study assignments (ISA), problems from TA sessions, and assigned sections from G. Strang's textbook. Problems are reproduced verbatim; solutions are complete, step-by-step, with verifications.

Week	Linear Algebra I
1	Vectors; linear combinations; span; linear independence
2	Dot product, angles, projections; cross product, areas; matrices: operations
3	Matrix algebra; transposition; matrix multiplication; determinants; scalar triple product
4	Matrix rank; inverse matrix via Gauss-Jordan method

Notation. Matrices are written row-wise: [a b; c d] — first row a b, second row c d; column vector is written as [x; y], row vector as [x y]. Powers and indices are written using ^ and _: A^{-1}, A^{T}, a_{11}.

2 Week 1 — Vectors (Linear Algebra I)

2.1 Linear Algebra I: Week 1 Formula Catalog

Vectors are columns in R^n; the entire week 1 builds on addition, scalar multiplication, combinations, span and independence. $(\mathbf{u}+\mathbf{v})_i = u_i + v_i$, $(c\mathbf{u})_i = c\,u_i$ (component-wise operations) (24) $c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n$ = linear combination, (25) span$\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ = set of all such combinations, (26) $\mathbf{v}_1, \ldots, \mathbf{v}_n$ are independent $\Longleftrightarrow c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n = \mathbf{0} \Longrightarrow c_1 = \cdots = c_n = 0$, (27) q $\mathbf{x}\ \|\mathbf{x}\| = x_1^2 + \cdots + x_n^2$, $\mathbf{x}\hat{} = \|\mathbf{x}\|$ (unit vector) (28) $\mathbf{v}$ = L $\mathbf{u}$ vector$\mathbf{v}$ of length L along$\mathbf{u}$: $\|\mathbf{u}\|$, (29)

$$d(P, Q) = \|Q - P\|. \quad (30)$$

2.2 Linear Algebra I, week 1: ISA exercises — questions and complete solutions

[ ] [ ] −2 −1

Throughout the vectors section, vectors are columns; in part A$\mathbf{u}$ =,$\mathbf{v}$ =. 3 5

[ ] [ ] −2 −1

1. Let u = [−2; 3], v = [−1; 5]. Compute u + v, u − v, 2u, −3v, 2u − 3v.

[Source: ISA LA, week 1, exercise 1]

Solution. Component-wise by (24): $\mathbf{u} + \mathbf{v}$ = [−3; 8]; $\mathbf{u} - \mathbf{v}$ = [−1; −2]; 2$\mathbf{u}$ = [−4; 6]; −3$\mathbf{v}$ = [3; −15]; 2$\mathbf{u}$ − 3$\mathbf{v}$ = [−4 + 3; 6 + 15] = [−1; 21].

2. Given a = −1, b = 2 ; find a + b, a − b, 3a + 2b. 4 −2

[5 -1; 6 12]

[Source: ISA LA, week 1, exercise 2]

Solution. a + b = □1□; **a − b** = □−3□; 3**a** + 2**b** = □−3 + 4□ = □ 1 □.

[2 6 12; -4 8] [2; 7]

3. Find x, if x + □−4□ = □ 3 □. 1 −2 □ □ 7 − 2

[Source: ISA LA, week 1, exercise 3]

Solution. Subtract the known vector from both sides: **x** = □ 3 + 4 □ = −2 − 1 □ □ 5 □ 7 □. −3

[x; y; z] - [2; -4; 1] = [8; 3; -2]

4. Find x, if 3x − 2u = v, where u = [−2; 1], v = [8; 2].

[Source: ISA LA, week 1, exercise 4]

Solution. Isolate: 3**x** = **v** + 2**u** = [8 − 4; 2 + 2] = [4; 4], hence **x** = [4/3; 4/3]. Scalar multiplication is distributive over addition, so dividing a vector by 3 means dividing each component by 3.

5. Compute the Euclidean length of the vectors u =, v = □−3□, w = □0□. 4 6 0

[Source: ISA LA, week 1, exercise 5]

Solution. By (28): ‖**u**‖ = √9 + 16 = 5; ‖**v**‖ = √4 + 9 + 36 = 7; ‖**w**‖ = 0 (only the zero vector has length zero).

□−3□

6. Find the unit vector in the same direction as v =, and verify that 4 its length is 1.

[−4/5; 3/5]

[Source: ISA LA, week 1, exercise 6]

Solution. ‖**v**‖ = 5, therefore ˆ**v** = **v**/5 =. Verification: ‖**v**ˆ‖ = 9/25 + 16/25 = 4/5 p 25/25 = 1. Division by the norm preserves direction (the scalar is positive) and makes the length equal to 1. □−4□

7. Find the vector of length 10 in the same direction as v =. 3 □−8□

[Source: ISA LA, week 1, exercise 7]

Solution. Here ‖**v**‖ = 5, so we scale by 10/5 = 2: the answer is 2**v** =

[6; -16]

indeed ‖2**v**‖ = 2 · 5 = 10 by the homogeneity of the norm.

8. Using vector subtraction and length, find the distance between P = (1, −2, 3) and Q = (5, 1, −3). □ □ 4 □, therefore d(P, Q) = ‖PQ‖ = √16 + 9 + 36 = √

[Source: ISA LA, week 1, ex. 8]

Solution. PQ = Q − P = □ 4 3 −6

[5-1; 1-(-2); -3-3]

= □ 4 3 −6

[4; 3; -6]

. Therefore d(P, Q) = ‖**PQ**‖ = $\sqrt{4^2 + 3^2 + (-6)^2}$ = √16 + 9 + 36 = √61.

9. Let$\mathbf{v}_1$ =,$\mathbf{v}_2$ =. Compute $2\mathbf{v}_1$ +$\mathbf{v}_2$, −$\mathbf{v}_1$ + 3$\mathbf{v}_2$, 4$\mathbf{v}_1$ − 2$\mathbf{v}_2$, 0$\mathbf{v}_1$ + 5$\mathbf{v}_2$. 2 3

[] [] [] [] [] 1 −1 − 1 −2 4 + 2 6

[Source: ISA LA, week 1, ex. 9]

Solution. $2\mathbf{v}_1$ +$\mathbf{v}_2$ =

[2 2; 6 3]

; −$\mathbf{v}_1$ + 3$\mathbf{v}_2$ =

[-1 3; -3 9]

; 4$\mathbf{v}_1$ − 2$\mathbf{v}_2$ =

[4 -2; 12 -6]

; 0$\mathbf{v}_1$ + 5$\mathbf{v}_2$ = (the zero scalar completely eliminates$\mathbf{v}_1$).

[0 5; 0 15]

5$\mathbf{v}_2$ =

[1 -1 7; -1 -2 4] + [2 6; 0 0]

.

10. Find c_1, c_2 such that c_1 + c_2 =; check by substitution. 1 2 1

[Source: ISA LA, week 1, ex. 10]

Solution. Component-wise equations: $c_1 - c_2 = 7$ and $c_1 + 2c_2 = 1$. Subtracting the first from the second:

$$c_1 + c_2 = 1;\ c_1 - c_2 = 7;\ 2c_1 + c_2 = 1.$$

$3c_2 = -6$, so $c_2 = -2$ and $c_1 = 5$. Verification:

$$[5 - 2;\ 5 + 2] = [7;\ 1]$$

.

11. Determine if$\mathbf{b}$ = □1□ can be written as$\mathbf{b}$ = c_1 □0□ + c_2 □1□ + c_3 □1□. 4 1 1 0

[Source: ISA LA, week 1, ex. 11]

Solution. Components give $c_1 + c_3 = 5$, $c_2 + c_3 = 1$, $c_1 + c_2 = 4$. Adding the first two and subtracting the third: $2c_3 = 2$, so $c_3 = 1$, $c_1 = 4$, $c_2 = 0$. The combination exists:$\mathbf{b}$ = 4$\mathbf{v}_1$ +0·$\mathbf{v}_2$ +1·$\mathbf{v}_3$. The uniqueness of the solution also shows that the three vectors are independent.

□−2□

12. Describe geometrically all vectors c·[−2; 3], c ∈ R; do the vectors [4; −6], [−2; 3], [6; −8] belong to this span?

[Source: ISA LA, week 1, ex. 12]

Solution. The span is the entire line through the origin in the direction (−2, 3). Checking: [4; −6] — yes (c = −2: −2·(−2) = 4 and −2·3 = −6); [−2; 3] — yes (c = 1); for [6; −8] we need −2c = 6 and 3c = −8, i.e. c = −3 and c = −8/3 — contradiction, so it does not lie on the line.

13. Are$\mathbf{v}_1$ =,$\mathbf{v}_2$ = linearly independent? Find non-zero c_1, c_2 with $c_1\mathbf{v}_1$ + 2 6 $c_2\mathbf{v}_2$ =0 and explain geometrically.

[Source: ISA LA, week 1, ex. 13]

Solution. $v_2 = 3v_1$, so $3v_1 - v_2 = 0$ — a non-trivial dependence ($c_1 = 3$, $c_2 = -1$). The vectors are dependent. Geometrically, both lie on the same line through the origin, so together they still only span this line — dependence means that one vector does not give a new direction.

[1 0; 1]

14. Arev$_1$ = □0□,v$_2$ = □1□,v$_3$ = □1□ linearly independent? If dependent, 1 1 2 express one in terms of a combination of the others.

[Source: ISA LA, week 1, ex. 14]

Solution. Assume$\mathbf{v}_3 = \alpha\mathbf{v}_1 + \beta\mathbf{v}_2$: components give α = 1 (first), β = 1 (second), α + β = 2 (third) — consistent. So$\mathbf{v}_3 = \mathbf{v}_1 + \mathbf{v}_2$, a non-trivial dependence exists, and the set is dependent. Only two independent directions are present; the span is a plane.

[1; 1; 1]

15. Letv$_1$ =,v$_2$ =. (1) Do they span R^2? (2) Forb = find

2 1 [] b$_2$ 7

c$_1$, c$_2$ with c$_1$v$_1$ + c$_2$v$_2$ =b. (3) Express by formula. 4

[Source: ISA LA, week 1, ex. 15]

Solution. (1)–(2) Solve $c_1 + 3c_2 = b_1$, $2c_1 + c_2 = b_2$. Eliminating: $c_1 = b_1 - 3c_2$, then $2b_1 - 6c_2 + c_2 = b_2$, so $c_2 = (2b_1 - b_2)/5$ and $c_1 = (-b_1 + 3b_2)/5$. The formulas are defined for any (b_1, b_2), so yes, the two vectors span R^2. (3) $c_1 = (-7 + 12)/5 = 1$, $c_2 = (14 - 4)/5 = 2$: $b = v_1 + 2v_2$.

Q = uv = (2: a = V) + 2Vo.

16. Letu$_1$ =,u$_2$ =. (1) Do they span R^2? (2) Describe 2 3

(1) u$_2$ = 6 u$_1$, accor

span{u$_1$,u$_2$} geometrically; (3) do they belong to it,? −4 2

[Source: ISA LA, week 1, ex. 16]

Solution. Comparing the ratios −6/−1 = 6 and 3/2 shows that the printed pair is not a multiple, so as printed it spans R^2; □−3□ in the standard version of the exercise$\mathbf{u}_2 = 3\mathbf{u}_1$ = — we solve that. (1)$\mathbf{u}_2 = 6\ 3\mathbf{u}_1$, so the span is only span{$\mathbf{u}_1$}. (2) Geometrically, this is the only line through the origin in the direction (−1, 2); two collinear □ 8 □ vectors never span a plane. (3) = $-8\mathbf{u}_1$? Check: −4 □1□ −8 · (−1) = 8, −8 · 2 = −16≠ −4, so no; for: need −c = 1 and 2c = 2 2 — c = −1 and c = 1, contradiction, so no (the point (1, 2) does not lie on the line y = −2x).

y = (1/2) [1 0; 1 1] x + [1; 0] (16)

17. Let v$_1$ = □0□,v$_2$ = □1□,v$_3$ = □1□. (1) Solve c$_1$v$_1$ + c$_2$v$_2$ + c$_3$v$_3$ =b for 1 1 0 □ □

c$_1$ = (b$_1$ + b$_2$ - b$_3$)/2

arbitraryb = □b$_2$□; (2) explain why the three vectors span R^3;

c$_1$ = (b$_1$ + b$_2$ - b$_3$)/2

□2□. (3) express 6

[Source: ISA LA, week 1, ex. 17]

Solution. (1) Components: $c_1 + c_3 = b_1$, $c_2 + c_3 = b_2$, $c_1 + c_2 = b_3$. Adding all three: $2(c_1 + c_2 + c_3) = b_1 + b_2 + b_3$, so $c_1 + c_2 + c_3 = \frac{b1+b2+b3}{2}$. Then $c_1 = \frac{b1-b2+b3}{2}$, $c_2 = \frac{-b1+b2+b3}{2}$, $c_3 = \frac{b1+b2-b3}{2}$. (2) A solution exists and is unique for any right-hand side, so the span is all of R^3 (and the vectors are independent). (3) $c_1 = \frac{4-2+6}{2} = 4$, $c_2 = \frac{-4+2+6}{2} = 2$, $c_3 = \frac{4+2-6}{2} = 0$: $\mathbf{b} = 4\mathbf{v}_1 + 2\mathbf{v}_2$.

$$b = 4v_1 + 2v_2$$

18. Three forces act on a part: $\mathbf{F_1}$ = N,$\mathbf{F_2}$ = N,$\mathbf{F_3}$ =

[10; 25; −5]

N. (1) Find the resultant force; (2) find $\mathbf{F_4}$ that brings the part into equilibrium: $\mathbf{F_1}$ +$\mathbf{F_2}$ +$\mathbf{F_3}$ +$\mathbf{F_4}$ =0.

[40; -15; -20]

[Source: ISA Linear Algebra, week 1, ex. 18]

Solution. (1) We add component-wise: $\mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 =$

begin{bmatrix} 10 25 -5 end{bmatrix}

= 50 N. (2) For equilibrium, we need $\mathbf{F}_4 = -(\mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3) =$

begin{bmatrix} -40 15 20 end{bmatrix}

N: exactly the opposite vector to the resultant force, so that the total force becomes zero.

2.3 Linear Algebra I, week 1: TA session problems (alternative set) — questions and complete solutions

The TA set contains eight short problems, each with an official solution; we present both, elaborating on the reasoning in detail.

1. Let a = (−1, 3, 2), b = (4, 0, −2). Compute a +b and 3a − b; find the midpoint between a and b.

[Source: TA Problems, Week 1, Problem 1]

Solution. a +**b** = (3, 3, 0); 3**a** − **b** = (−3 − 4, 9 − 0, 6 + 2) = (−7, 9, 8). The midpoint is the average $\frac{a+b}{2}$ = ($\frac{3}{2}$, $\frac{3}{2}$, 0) — the same averaging idea as in the scalar formula for the midpoint of a segment.

2. Givenp +q = (5, 7) andp − q = (−1, 3); findp andq.

[Source: TA Problems, Week 1, Problem 2]

Solution. Add the two equations: 2**p** = (4, 10), so**p** = (2, 5). Subtract: 2**q** = (6, 4), so**q** = (3, 2). Check: (2, 5) + (3, 2) = (5, 7) and (2, 5) − (3, 2) = (−1, 3). The "add and subtract" method is the vector analog of simultaneously solving two scalar equations.

3. Letu = (1, 2, −1),v = (2, −1, 0). Compute −u + 3v; write cu + dv component-wise; can cu + dv equal (5, 0, −1)?

[Source: TA Problems, Week 1, Problem 3]

Solution. $-\mathbf{u} + 3\mathbf{v} = (-1, -2, 1) + (6, -3, 0) = (5, -5, 1)$. Component-wise $c\mathbf{u} + d\mathbf{v} = (c + 2d, 2c - d, -c)$. To get $(5, 0, -1)$: the third component forces $c = 1$; then $c + 2d = 5$ gives $d = 2$; and indeed $2c - d = 0$. Answer: yes, $\mathbf{u} + 2\mathbf{v} = (5, 0, -1)$.

4. Let $\mathbf{r} = (2, -1), \mathbf{s} = (-6, 3)$. Describe all combinations $c\mathbf{r} + d\mathbf{s}$; do the vectors span $\mathbb{R}^2$? Find the dependence relation.

[Source: TA Problems, Week 1, Problem 4]

Solution. Since $\mathbf{s} = -3\mathbf{r}$, any combination equals $(c - 3d)\mathbf{r}$ — always a multiple of $\mathbf{r}$. The span is only the line through $\mathbf{r}$: the vectors do not span a plane. The relation $3\mathbf{r} + \mathbf{s} = \mathbf{0}$ is nontrivial, which proves dependence (cf. (27)).

5. Let $\mathbf{v} = (1, 0, 1), \mathbf{w} = (1, 2, 0)$. Write $c\mathbf{v} + d\mathbf{w} = (x, y, z)$; find one equation relating x, y, z; check $\mathbf{b} = (4, 6, 1)$.

[Source: TA Problems, Week 1, Problem 5]

Solution. $c\mathbf{v} + d\mathbf{w} = (c + d, 2d, c)$. Reading: $z = c$ and $y = 2d$, so $x = z + \frac{y}{2}$ — the only linear equation describing the plane. For $\mathbf{b}$: right side $= 1 + \frac{6}{2} = 4 = x$, so $\mathbf{b}$ lies in the plane; from $z = 1$, $y = 6$: $c = 1$, $d = 3$.

6. Find c, d such that $c(1, 2) + d(3, -1) = (7, 3)$; write and solve the component-wise equations.

[Source: TA Problems, Week 1, Problem 6]

Solution. Components: $c + 3d = 7$ and $2c - d = 3$. From the second $d = 2c - 3$; substituting: $c+6c-9 = 7$, so $7c = 16$, $c = \frac{16}{7}$, $d = 32_7 - 3 = 11_7$. Both components are verified exactly.

7. Let $\mathbf{u} = (1, 1, 0)$, $\mathbf{v} = (0, 1, 1)$, $\mathbf{w} = (1, 2, 1)$. Does $\mathbf{w}$ add a new direction? Describe span{u,v,w}; give a relation equal to zero.

[Source: TA Problems, Week 1, Problem 7]

Solution. $\mathbf{w} = \mathbf{u} + \mathbf{v}$ (component-wise check: $1 + 0 = 1$, $1 + 1 = 2$, $0 + 1 = 1$), so $\mathbf{w}$ is redundant. The relation $\mathbf{u} + \mathbf{v} - \mathbf{w} = \mathbf{0}$ is non-trivial, so the three vectors are dependent, and the span is just the plane spanned by $\mathbf{u}$ and $\mathbf{v}$ (they are not scalar multiples).

8. Can two vectors span all of $\mathbb{R}^3$? Can three vectors span $\mathbb{R}^3$ and at the same time be dependent? Explain through "new directions."

[Source: TA Problems, Week 1, Problem 8]

Solution. Two vectors span at most a plane — each vector gives at most one new direction, and two directions cannot fill three-dimensional space. Three dependent vectors also span at most a plane: dependence means one of them is a combination of the others and does not add a third direction. To achieve $\mathbb{R}^3$, three independent directions are needed; therefore, three dependent vectors will never span $\mathbb{R}^3$.

3 Week 2 — Scalar and Vector Products, Introduction to Matrices (Linear Algebra I)

3.1 Linear Algebra I, Week 2: Formula Catalog

Week 2 adds two products of vector geometry and the first matrix concepts. **u** ·**v** **u** ·**v** = $u_1v_1 + \cdots + u_nv_n = \mathbf{u}^T\mathbf{v}$, cos θ = ‖**u**‖‖**v**‖, (45) $\text{proj}_{\mathbf{v}}\mathbf{u} = \mathbf{u}\cdot\mathbf{v}$ **u** ⊥ **v** ⇐⇒ **u** ·**v** = 0, **v** ·**vv**, (46) □ □

$$[u_2v_3 - u_3v_2]$$

u ×**v** = □$u_3v_1 - u_1v_3$□, ‖**u** ×**v**‖ = ‖**u**‖‖**v**‖| sin θ| (area),

$$u_1v_2 - u_2v_1$$

(47) **u** ×**v** = −(**v** ×**u**), (**u** ×**v**) ⊥ **u**,**v**, (48)

$$A = [a_{ij}] \in Rm\times n,\ (A + B)_{ij} = a_{ij} + b_{ij},\ (cA)_{ij} = c\,a_{ij},$$

(49) A**x** = $x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$ (columns), (50) where $\mathbf{a}_j$ is column j of matrix A. Dimensions must match: A**x** exists if and only if A has size m×n and**x** ∈ R^n, and the result lies in R^m. Diagonal matrix D = diag(d_1, . . . , d_n) scales each coordinate: D**x** = $(d_1x_1, \ldots, d_nx_n)^T$.

3.2 Linear Algebra I, Week 2: Lecture Classroom Problems — Questions and Full Solutions

[3] [] 4

1. Letu =,v =. Computeu ·v; classify the angle. −2 5

[Source: LA Lecture, Week 2, Problem 1]

Solution. **u** ·**v** = 3 · 4 + (−2) · 5 = 12 − 10 = 2 > 0. Positive scalar product means cos θ > 0, so the angle is acute. □ □ □ □ 2 −1

2. With u = [2; −1; 2], v = [−1; 3; 0]: find ‖u‖ and dist(u, v).

[Source: LA Lecture, Week 2, Problem 2]

Solution. ‖**u**‖ = sqrt(4 + 1 + 4) = 3. **u** − **v** = [3; −4; 2], so dist(**u**, **v**) = sqrt(9 + 16 + 4) = sqrt(29).

3. Find k so thatu = □ k □ is orthogonal tov = □1□. −1 4

[Source: LA Lecture, Week 2, Problem 3]

Solution. Orthogonality means**u** ·**v** = 0: 6 + k − 4 = 0, so k = −2.

Pemenne. OpToroHasIbHOCTb — 3TO0 U·V=0: 6+k−4=0, 3HauUT k = −2.

4. Find the angle between u = and v =. 1 −1

[Source: LA Lecture, week 2, problem 4]

Solution. **u** ·**v** = 1 − 1 = 0: the vectors are perpendicular, θ = 90° — no formulas needed, as soon as the dot product becomes zero.

$$\cos\varphi = (u\cdot v) / (||u||\ ||v||)$$

5. Project u = onto v =. 2 1 □3□

[Source: LA Lecture, week 2, problem 5]

Solution. $\mathbf{u}\cdot\mathbf{v} = 6, \mathbf{v}\cdot\mathbf{v} = 2$, therefore $\text{proj}_{\mathbf{v}}\mathbf{u} = \frac{6}{2}\mathbf{v} = 3\mathbf{v}$ =. Remainder 3 □ 1 □ **u** − 3**v** = satisfies (**u** − 3**v**) ·**v** = 0: decomposition into parallel and −1 + perpendicular components. □ □ □ □ 1 3

6. Compute u ×v for u = □2□,v = □−1□ and check orthogonality. 0 2 □ □ □ □

$$2 \cdot 2 - 0 \cdot (-1)\ [4]$$

[Source: LA Lecture, week 2, problem 6]

Solution. By (47):**u** ×**v** = □ 0 · 3 − 1 · 2 □ = □−2□. Check: 1 · 4 + 2 · (−2) + 0 ·

$$u \times v = [2; 0; 2] \times [-1; 2; 0] = [0 \cdot 0 - 2 \cdot 2; 2 \cdot (-1) - 2 \cdot 0; 2 \cdot 2 - 0 \cdot (-1)] = [-4; -2; 4]\ (16)$$

(−7) = 0 and 3 · 4 + (−1)(−2) + 2 · (−7) = 0 — orthogonal to both inputs. □ □ □ □ 2 1

7. Find the area of the parallelogram with sides a = □0□,b = □ 3 □. 1 −1 □ □ □ □

$$0 \cdot (-1) - 1 \cdot 3 - 3$$

[Source: LA Lecture, week 2, problem 7]

Solution. a ×**b** = □1 · 1 − 2 · (−1)□ = □ 3 □, so area ‖**a** ×**b**‖ =

$$2 \cdot 3 - 0 \cdot 1 = 6$$

√9 + 9 + 36 = √ √ 54 = 3 6. □2 □ −1 4

8. For A =: specify the size, find a_{13}, a_{22} and write the columns 0 3 5 $\mathbf{a}_1$,$\mathbf{a}_3$.

[Source: LA Lecture, week 2, problem 8]

Solution. A has size 2 × 3 (rows first). a_{13} = 4 (row 1, column

$$A = [2\ 4; 0\ 3\ 5]$$

3), a_{22} = 3.$\mathbf{a}_1$ =,$\mathbf{a}_3$ =. 0 5

$$A = [1\ -2; 0\ 3],\ R = [4; 1],\ H = (94 - R^T B).$$

9. Let A = [1 −2; 3 0], B = [4 1; −1 5]. Find 2A − B.

[Source: LA Lecture, week 2, problem 9]

Solution. 2A = [2 −4; 6 0]; subtract element-wise: 2A − B = [2 − 4, −4 − 1; 6 − (−1), 0 − 5] = [−2 −5; 7 −5].

10. A has size 3 × 2, B is 2 × 3,$\mathbf{x} \in \mathbb{R}^2$,$\mathbf{y} \in \mathbb{R}^3$. Which of Ax, Ay, Bx, By are defined?

[Source: LA Lecture, week 2, problem 10]

Solution. A**x**: (3 × 2)(2 × 1) OK, result in $\mathbb{R}^3$. B**y**: (2 × 3)(3 × 1) OK, result in $\mathbb{R}^2$. A**y** is inconsistent (2 ≠ 3) and B**x** is inconsistent (3 ≠ 2): both undefined. Only inner dimensions must match. □ □ 4

11. Let D = diag(2, −1, 3),x = □ 5 □. Find Dx and I_3x. −2 □ □ □ □ 2 · 4 8

[Source: LA Lecture, week 2, problem 11]

Solution. D**x** = □ −1 · 5 □ = □−5□: each coordinate is scaled

$$3 \cdot (-2) = -6$$

. I_3**x** =**x** = □ 5 □: the identity matrix does nothing. −2

12. Let A = □−1 3□,x =. Compute Ax by the column picture and through −1 4 0 scalar products of rows.

[1 2 0; -1 3 4; 0 0 8]

[Source: LA Lecture, week 2, problem 12]

Solution. Columns: 2 □−1□+(−1) □3□ = □−5□. Rows: [1 2]·**x** = 0, [−1 3]·**x** = −5, 4 0 8 □ □ 0 [4 0]·**x** = 8, stacking we get the same □−5□. Both pictures always agree; 8 the column view explains what the product means.

13. Linear transformation L: $R^3 \to R^3$ is defined by L(x) = Ax, where A = [−1 3 0; 2 −1 4; 0 1 2].

13. Given A = and Ax =; find x. 3 −1 5

[Source: LA Lecture, week 2, problem 13]

Solution. Components: $x_1 + 2x_2 = 7$, $3x_1 - x_2 = 5$. From the second $x_2 = 3x_1 - 5$;

x = [17/7]

substitute: $x_1 + 6x_1 - 10 = 7$, so $7x_1 = 17$, $x_1 = {}^{17}_{7}$, $x_2 = 16_7$. So **x** =. 16/7

14. Non-zero vectors u +v and u − v are orthogonal. What follows about ‖u‖ and ‖v‖?

[Source: LA Lecture, week 2, problem 14]

Solution. Orthogonality gives 0 = (**u**+**v**)·(**u**−**v**) = $\|\mathbf{u}\|^2 - \|\mathbf{v}\|^2$ (cross terms cancel), so ‖**u**‖ = ‖**v**‖: equal lengths, and **u** and **v** are half-diagonals of a rectangle.

3.3 Linear Algebra I, week 2: ISA exercises — questions and full solutions

1. Let u = $(-3, 2)^{\mathbf{T}}$,v = $(4, 5)^{\mathbf{T}}$. Compute u·v; is the angle acute, right, or obtuse?

[Source: LA ISA, week 2, ex. 1]

Solution. u ·**v** = −12 + 10 = −2 < 0, so cos θ < 0: the angle is obtuse.

2. Let a = $(1, -2, 4)^{\mathbf{T}}$, b = $(3, 0, -1)^{\mathbf{T}}$. Find a · b, ‖a‖, ‖b‖, ‖a − b‖.

[Source: ISA LA, week 2, ex. 2]

Solution. a · **b** = 3 + 0 − 4 = −1; ‖**a**‖ = √1 + 4 + 16 = √21; ‖**b**‖ = √9 + 0 + 1 = √ 10; **a** − **b** = $(-2, -2, 5)^{\mathrm{T}}$, ‖**a** − **b**‖ = √4 + 4 + 25 = √ 33.

3. Find k such that u = $(2, k, 3)^{\mathbf{T}}$ is orthogonal to v = $(-1, 4, 2)^{\mathbf{T}}$.

[Source: ISA LA, week 2, ex. 3]

Solution. u · **v** = −2 + 4k + 6 = 0 ⇒ 4k = −4 ⇒ k = −1.

4. Find the exact cos θ, then θ to the nearest degree for u = $(3, 1)^{\mathbf{T}}$, v = $(-2, 4)^{\mathbf{T}}$. √ √ √

[Source: ISA LA, week 2, ex. 4]

Solution. u · **v** = −6+4 = −2; ‖**u**‖ = 10, ‖**v**‖ = 20 = 2 5; cos θ = ${}^{-2}_{5}$ = ${}^{-2}_{2}$ = √ √ √

u·v = 3·(−2) + 1·4 = −6+4 = −2; ||u|| = √10, ||v|| = √20 = 2√5; cosθ = (u·v)/(||u||·||v||) = (−2)/(√10·2√5) = (−1)/(√50) = (−1)/(5√2) = −√2/10

$-{}^{1}_{2} \approx -0.1414$, therefore θ ≈ 98°. √ 5

5. Find the projection of u = (6, 1)^{T} onto v = (2, −1)^{T}; verify that u − proj is orthogonal to v.

[Source: ISA LA, week 2, ex. 5]

Solution. **u** · **v** = 12 − 1 = 11, **v** · **v** = 5: proj$_{v}$ **u** = $^{11}_{5}$ (2, −1)^{T} = (22$_{5}$, − $^{11}_{5}$)^{T} . Remainder: **u** − proj = ($^{8}_{5}$, 16$_{5}$)^{T} ; dot product with **v**: 16$_{5}$ − $^{16}_{5}$ = 0. Verified.

6. Nonzero p, q satisfy p · q = ‖p‖‖q‖. What can you conclude about their directions?

[Source: ISA LA, week 2, ex. 6]

Solution. From (45), cos θ = 1, thus θ = 0: the vectors are directed exactly the same (one is a positive multiple of the other). Equality in the angle formula is a limiting case of the Cauchy–Bunyakovsky–Schwarz idea.

7. For a = (2, 1, −1)^{T}, b = (0, 3, 2)^{T} compute a×b and b×a; verify the sign relationship.

$$[2 \cdot 3 - (-1) \cdot 0; 5 -5]$$

[Source: ISA LA, week 2, ex. 7]

Solution. **a** × **b** = □(−1) · 0 − 2 · 2□ = □−4□, and **b** × **a** = □ 4 □ = − (**a** × **b**), which

$$[2·3 - 1·0; 1·0 - (-1)·3; (-1)·3 - 2·2] = [6; 3; -7]$$

corresponds to (48).

8. Verify that a × b from exercise 7 is orthogonal to both a and b.

[Source: ISA LA, week 2, ex. 8]

Solution. **a** · (**a** × **b**) = 2 · 5 + 1 · (−4) + (−1) · 6 = 10 − 4 − 6 = 0; **b** · (**a** × **b**) = 0 · 5 + 3 · (−4) + 2 · 6 = −12 + 12 = 0. Both are zero, as guaranteed after (47).

9. Find the area of the parallelogram with sides u = (1, 2, 2)^{T}, v = (3, −1, 0)^{T}. □ □ □ □

$$[2 \cdot 0 - 2 \cdot (-1); 2 \cdot 3 - 1·0; 1 \cdot (-1) - 2 \cdot 3]$$

□; area = √4 + 36 + 49 = √

[Source: ISA LA, week 2, ex. 9]

Solution. **u** × **v** = □ 2 · 3 − 1 · 0 □ = □ 6 89.

$$||u \times v|| = √((2·0 - 2·(-1))^2 + (2·3 - 1·0)^2 + (1·(-1) - 2·3)^2) = √(4 + 36 + 49) = √89$$

10. P = (1, 0, 1), Q = (3, 1, 0), R = (0, 2, 4): find the area of triangle PQR.

[Source: ISA LA, week 2, ex. 10]

Solution. Edges from P: **PQ** = (2, 1, −1)^{T}, **PR** = (−1, 2, 3)^{T}. Cross product: **PQ** × **PR** = (1·3 − (−1)·2; (−1)·(−1) − 2·3; 2·2 − 1·(−1)) = (5, −5, 5), norm

$$sqrt(25 + 25 + 25) = sqrt(75) = 5sqrt(3)$$

; triangle area = (1/2)·5√3 = 5√3/2 (half the parallelogram area). Swapping **u** and **v** changes the sign of the cross product, area doesn't change.

11. Find all real t for which a = (1, t, 2)^{T} and b = (3, 6, 6)^{T} satisfy a × b = 0; if none, explain.

[Source: ISA LA, week 2, ex. 11]

Solution. Parallel vectors give zero cross product, so we need **a** = c**b**: 1 = 3c ⇒ c = $^{1}_{3}$; then t = 6c = 2; check third: 2 = 6c = 2 — consistent. So t = 2 is the only value (then **a** = $^{1}_{3}$ **b**).

12. Let a = (1, 0, 2)^{T}, b = (2, 1, −1)^{T}, c = (0, 3, 1)^{T}. Compute a · (b × c); interpret the zero value. □ □ □ □

1 · 1 − (−1) · 3 4

[Source: ISA LA, week 2, exercise 12]

Solution. **b** × **c** = □(−1) · 0 − 2 · 1□ = □−2□; **a** · (**b** × **c**) = 4 + 0 + 12 = 16.

[1 0 2; 2 1 -1; 0 3 1]

This is the signed volume of the parallelepiped spanned by three vectors; zero value would mean coplanarity (no 3D volume). Here |16| = 16 cubic units. □2 □ −1 0

13. Let A = [2 −1 0; 4 3 5]. Specify the size; find a_{12}, a_{23}; write each row and column as a vector.

[Source: ISA LA, week 2, exercise 13]

Solution. Size 2 × 3 (rows first). a_{12} = −1 (row 1, column 2), a_{23} = 5. Rows: [2 −1 0], [4 3 5]; columns: [2; 4], [−1; 3], [0; 5].

[1] [ ] 3 5 −1

14. Let A = [1 3; −2 4], B = [5 −1; 0 2]. Compute A + B, A − B, −2A, 3A − 2B.

[Source: ISA LA, week 2, exercise 14]

Solution. A + B = [6 2; −2 6]; A − B = [−4 4; −2 2]; −2A = [−2 −6; 4 −8]; 3A − 2B = [3 − 10, 9 + 2; −6 − 0, 12 − 4] = [−7 11; −6 8].

15. A has size 4 × 3, B — 3 × 2,x ∈ R^{2},y ∈ R^{3},z ∈ R^{4}. Which of Ax, Ay, Az, Bx, By, Bz are defined and with what output dimensions?

[Source: ISA LA, week 2, exercise 15]

Solution. A**y**: (4 × 3)(3 × 1) — defined, result R^{4}. B**x**: (3 × 2)(2 × 1) — defined, result R^{3}. All others (A**x**, A**z**, B**y**, B**z**) have mismatched inner dimensions and are undefined.

16. Let D = diag(−2, 3, 1),x = (4, −1, 5)^{T}. Compute Dx; describe what D does to each coordinate.

Dx = [−2 0 0; 0 3 0; 0 0 1], x = (2, −1, 3)^{T}.

[Source: ISA LA, week 2, exercise 16]

Solution. D**x** = first coordinate scaled by −2 (stretching with reflection), second — by 3, third — by 1 (unchanged). Diagonal matrix — three independent scalings of coordinate axes. □ 1 □ −2 3 ,**x** = (2, −1, 3)^{T}.

17. Let A = Compute Ax (a) as a combination of 4 0 −1 columns, (b) through scalar products of rows.

1 −2 2 + 2 + 9 = 13

[Source: ISA LA, week 2, exercise 17]

Solution. (a) 2 − 1 + 3 =

2 [1; 2; 3] = [2; 4; 6] − [1; 2; 0] + [3; 0; 3] = [4; 2; 9]

. (b) Row 1: 1 · 2 + (−2)(−1) + 3 · 3 = 13; row 2: 4 · 2 + 0 · (−1) + (−1) · 3 = 5; row 3: 2 · 2 + 2 · (−1) + 2 · 3 = 6 − 2 + 6 = 10.

[1 4 2; −2 0 2; −1 −1 2] [2; −1; 3] = [13; 5; 10]

Same result, as expected.

18. Let $\mathbf{a}_1 = (1, 2)^T$, $\mathbf{a}_2 = (3, -1)^T$. Find $\mathbf{x} = (x_1, x_2)^T$ with $x_1\mathbf{a}_1 + x_2\mathbf{a}_2 = (11, 1)^T$.

[Source: ISA LA, week 2, exercise 18]

Solution. Components: $x_1 + 3x_2 = 11$, $2x_1 - x_2 = 1$. From the second $x_2 = 2x_1 - 1$; then $x_1 + 6x_1 - 3 = 11 \Rightarrow 7x_1 = 14 \Rightarrow x_1 = 2$, $x_2 = 3$. □ 2 □ 4

19. Let A = [2 4; −1 −2]. Describe all x with Ax = 0; why are the columns dependent?

[Source: ISA LA, week 2, exercise 19]

Solution. $2x_1 + 4x_2 = 0$ and $-x_1 - 2x_2 = 0$ are the same equation: $2x_1 = -4x_2$. Solutions: $\mathbf{x} = t\cdot[-2; 1]$, $t \in R$ — the entire solution line. There is a nontrivial combination $-2\cdot\mathbf{a}_1 + 1\cdot\mathbf{a}_2 = \mathbf{0}$, which is the linear dependence of the columns (column 2 equals −2 times column 1).

[] [] 1 = (2, −1, 4)$_T$ and A 0 = (3, 2, 0)$_T$. Write A

20. A has size 3 × 2, with A 0 1 □−2□ and compute A without scalar products of rows. 5 □ □

A = [2 3; -1 2; 5 4]

[Source: ISA LA, week 2, exercise 20]

Solution. The two images are the columns of A: A = □−1 2□. Linearity: A = 5 4 0

[2 3; 5 11]

$-2\,A\mathbf{e}_1 + 5\,A\mathbf{e}_2 = -2$ □−1□ + 5 □2□ = □ 12 □. 4 0 −8

21. Expand (u +v) · (u − v) for vectors of the same dimension; prove that u +v ⊥ u − v if and only if ‖u‖ = ‖v‖.

[Source: ISA LA, week 2, exercise 21]

Solution. Distributing: $(\mathbf{u} + \mathbf{v}) \cdot (\mathbf{u} - \mathbf{v}) = \mathbf{u}\cdot\mathbf{u} - \mathbf{u}\cdot\mathbf{v} + \mathbf{v}\cdot\mathbf{u} - \mathbf{v}\cdot\mathbf{v} = \|\mathbf{u}\|^2 - \|\mathbf{v}\|^2$ (middle terms cancel). Orthogonality means this equals zero, i.e. $\|\mathbf{u}\|^2 = \|\mathbf{v}\|^2$, and since norms are non-negative, $\|\mathbf{u}\| = \|\mathbf{v}\|$.

22. Self-check: (1) what object gives the scalar product, and what gives the vector product? (2) How does the sign ofu ·v classify the angle? (3) Why does a 2 × 3 matrix multiply a vector from R^3 but not from R^2? (4) How do components ofx act in Ax? (5) Which operations from week 2 require equal dimensions, and which require vectors specifically from R^3?

[Source: LA ISA, week 2, self-check]

Solution. (1) Scalar; vector (in R^3). (2) Positive ⇒ acute, zero ⇒ right, negative ⇒ obtuse. (3) The product A**x** requires matching inner dimensions: 2 × 3 times 3 × 1 works; times 2 × 1 — no. (4) Like weights in a linear combination of columns $x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$. (5) Dot product, addition and subtraction require equal dimensions; cross product and its geometry (areas, normals) exist only in R^3.

3.4 Linear Algebra I, week 2: TA session (15 problems) — questions and complete solutions

[] [] −2 5

1.a =,b =: computea ·b; classify the angle. 4 1

[Source: LA TA, week 2, problem 1]

Solution. a ·b = −10 + 4 = −6 < 0: obtuse. □ □ −2

2. Find the norm and unit vector in the direction ofv = □ 1 □. 2 □ □ −2/3

[Source: LA TA, week 2, problem 2]

Solution. ‖**v**‖ = √4 + 1 + 4 = 3; unit vector ˆ**v** = □ 1/3 □. 2/3

□ □ □ □ 1 −2

3. Find the distance betweenp = □−2□ andq = □ 2 □. 4 1 □ □ 3 □; distance = √9 + 16 + 9 = √

[Source: LA TA, week 2, problem 3]

Solution. p − q = □−4 34. 3 □ □ □ □ 1 2

4. Find t, if □t□ is orthogonal to □−3□. 2 4

[Source: LA TA, week 2, problem 4]

Solution. 2 − 3t + 8 = 0 ⇒ t = $^{10}_{3}$.

[2; -1] · [t; -3] = 0 ⇒ 2t + 3 = 0 ⇒ t = -3/2

5. Find cos θ foru =,v =. 1 3

cos θ = (1)/(50) = (1)/(5√10)

[Source: LA TA, week 2, problem 5]

Solution. u ·v = 1, ‖**u**‖ = 5, ‖**v**‖ = 10; cos θ = $^{1}_{50}$ = $^{2}_{10}$. √

6. Projectu = (5, −1)^{T} ontov = (2, 1)^{T}; writeu as parallel + orthogonal components.

[7/5; 18/5]

[Source: LA TA, week 2, problem 7]

Solution. u ·v = 9,**v ·v** = 5: proj =

(9/5)[2; 1]

. Orthogonal component:

[5; −1] − [14/5; 9/5]

u − proj =

[11/5; −18/5]

. Decomposition:

u = [14/5; 9/5] + [11/5; −18/5]

.

7. Computea ×b fora = □−1□,b = □4□. 3 1 □ □ □ □

$$(-1) \cdot 1 - 3 \cdot 4 - 13$$

[Source: LA TA, week 2, problem 8]

Solution. $\mathbf{a} \times \mathbf{b} = \begin{vmatrix} 3 \cdot 0 - 2 \cdot 1 \end{vmatrix} = \begin{vmatrix} -2 \end{vmatrix}$.

$$| -4 \; 8; 0 \; -1 |$$

8. A = (0, 0, 0), B = (2, 1, 0), C = (1, 0, 3): find the area of the triangle.

$$[2 \; 1 \; 0; 1 \; 0 \; 3]$$

[Source: LA TA, week 2, problem 9]

Solution. $\mathbf{AB} = \mathbf{AC}$ = cross product $\begin{vmatrix} \end{vmatrix} \begin{vmatrix} \end{vmatrix}$

$$[1 \; 0 \; 0; 0 \; 1 \; 0; 0 \; 0 \; 1]^{1/2}$$

$= \begin{vmatrix} 0 \cdot 1 - 2 \cdot 3 \end{vmatrix} = \begin{vmatrix} -6 \end{vmatrix}$, norm = 46; area = $\frac{46}{2}$.

$S = (1/2) \cdot || (2; 1; 0) \times (1; 0; 3) || = (1/2) \cdot || (3; -6; -1) || = (1/2) \cdot \sqrt{(3^2 + (-6)^2 + (-1)^2)} = (1/2) \cdot \sqrt{46}$

9. For M =: size, m_{21}, all columns. 4 3 5

$$[1 \; 0 \; -2; 4 \; 3 \; 5]$$

[Source: LA TA, week 2, problem 10]

Solution. 2×3; $m_{21} = 4$; columns,,. 4 3 5

$$[\; 2 \; 1 \; 0; 0 \; 4 \; 3; 5 \; 0 \; 0 \;]$$

10. Let A = [2 1; −1 3], B = [0 4; 2 −2]. Find 3A + 2B.

[Source: LA TA, week 2, problem 11]

Solution. 3A = [6 3; −3 9], 2B = [0 8; 4 −4]; sum = [6 11; 1 5].

11. Let A = [2 −1 3; 0 4 1], x = [1; −2; 3]. Compute Ax as a combination of columns.

[Source: LA TA, week 2, problem 12]

Solution. $1 - 2 + 3 = =$.

$$[0 \; 4 \; 1; -2 \; 0 \; 4; 1 \; 3 \; 2] \; [x; y; z] = x[0; -2; 1] + y[4; 0; 3] + z[1; 4; 2] = [0x + 4y + 1z; -2x + 0y + 4z; 1x + 3y + 2z] = [4y + z; -2x + 4z; x + 3y + 2z] = [9; 5; 13]$$

$$[2 \; -1; 3 \; 2] + [2; 2] + [9; 13]$$

12. Let$\mathbf{a}_1$ =,$\mathbf{a}_2$ =,b =. Find allx with $[\mathbf{a}_1 \mathbf{a}_2]\mathbf{x} = \mathbf{b}$. 2 4 6

[Source: LA TA, week 2, problem 13]

Solution. Both equations reduce to $x_1 + 2x_2 = 3$ (system is consistent but

$$[3 - 2t; -7 \; 2; -9 \; 4]$$

redundant, since$\mathbf{a}_2 = 2\mathbf{a}_1$ and$\mathbf{b} = 3\mathbf{a}_1$). General solution:$\mathbf{x} =$, $t \in R$ — t infinitely many weights for the same combination.

13. Without revealing coordinates, explain why $(\mathbf{u} \times \mathbf{v}) \cdot (\mathbf{u} + \mathbf{v}) = 0$.

[Source: LA TA, week 2, problem 14]

Solution. The cross product is orthogonal to both its inputs, so it's orthogonal to any of their combinations: $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{u} = 0$ and $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{v} = 0$; adding, the dot product with$\mathbf{u} + \mathbf{v}$ equals $0 + 0 = 0$.

$$[u_1 \; u_2 \; u_3] \times [v_1 \; v_2 \; v_3] \cdot ([u_1 \; u_2 \; u_3] + [v_1 \; v_2 \; v_3]) = 0$$

14. A has size 2 × 2, with A·[1; 0] = [2; −1] and A·[0; 1] = [3; 4]. Find A and A·[5; −2].

[Source: TA LA, week 2, problem 15]

Solution. The images of the standard basis vectors are the columns:

$$A = [2\ 3; -1\ 4]$$

. Then A·[5; −2] = 5·[2; −1] − 2·[3; 4] = [10 − 6; −5 − 8] = [4; −13]; linearity turns the computation into two scalings without scalar products.

4 Week 3 — Matrix algebra, determinants, mixed product (Linear Algebra I)

4.1 Linear Algebra I: formula catalog for week 3

$$(A + B)_{ij} = a_{ij} + b_{ij},\ (cA)_{ij} = c\,a_{ij},\ (A^T)_{ij} = a_{ji},\ (A^T)^T = A,\ (65)\ (A + B)^T = A^T + B^T,\ (cA)^T = cA^T,\ A^T = A\ (\text{symmetric}),$$

(66)

$$A \in R^{m\times n},\ B \in R^{n\times p} \Rightarrow AB \in R^{m\times p},\ (67)$$

(67)

$$(AB)_{ij} = \Sigma_{k=1}^{n}\, a_{ik}b_{kj},\ (68)$$

AB = [A$\mathbf{b}_1$ · · · A$\mathbf{b}_p$] (column rule), (71) $\mathbf{u}^T\mathbf{v}$ = scalar, $\mathbf{u}\mathbf{v}^T$ = matrix (outer product). (72) a b det = ad − bc (signed area), (73) c d $C_{ij} = (-1)^{i+j}$ det M_{ij} (cofactor),

$$C_{ij} = (-1)^{i+j}\ det\ M_{ij},\ det\ A = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13},\ (74)$$

where M_{ij} is the minor (row i and column j deleted). Expansion along any row or column is allowed; rows and columns with zeros reduce the arithmetic. For triangular matrices, the determinant equals the product of diagonal elements. Rules for elementary row operations are collected in table 1. Table 1: Determinant rules for square matrices (versions for rows; versions for columns are also valid).

Operation on A	**Effect on det A**
Swapping two rows Multiplying a row by c	changes sign multiplies determinant by c no change determinant equals 0 $\det(A^T) = \det A$
Adding a multiple of a row to another Zero row or two proportional Transposing Product of square matrices of the same size	det(AB) = det A det B

Scaling the entire n × n matrix scales all n rows: $\det(cA) = c^n \det A$. Determinants are not distributive with respect to addition: for A = B = I_2, det(A + B) = 4, while det A + det B = 2. (**u** ×**v**) ·**w** =**u** · (**v** ×**w**) = det[**u v w**] (scalar triple product), (75) h = |(**u** ×**v**) ·**w**| $V_{parallelepiped}$ = |(**u** ×**v**) ·**w**|,, (76) ‖**u** ×**v**‖ $V_{tetrahedron} = \frac{1}{6}$ | det[**u v w**]|, T = 0 ⇐⇒ **u**,**v**,**w** are coplanar. (77)

4.2 Linear Algebra I: lecture problems for week 3 (15) — questions and complete solutions

[] [2] 1 −2 1

1. Let A = [1 −2; 3 0], B = [2 1; −1 4]. Compute 2A − B and specify the (2, 1) element; can you add A to a 2 × 3 matrix?

[Source: LA Lecture, week 3, problem 1]

Solution. 2A = [2 −4; 6 0]; 2A − B = [2 − 2, −4 − 1; 6 − (−1), 0 − 4] = [0 −5; 7 −4]; the (2, 1) element is 2(3) − (−1) = 7. Addition with a 2 × 3 matrix is undefined — both dimensions must match, not just the total number of elements.

2. Let C = [1 2 1; 0 3 2]. Write C^T with its size; compute CC^T; is it symmetric?

[Source: LA Lecture, week 3, problem 2]

Solution. C^T = [1 0; 2 3; 1 2] (3 × 2). CC^T = [1 + 4 + 1, 0 + 6 − 2; 0 + 6 − 2, 0 + 9 + 4] = [6 4; 4 13]. Diagonal elements are rows, scalar multiplied by themselves; off-diagonal elements are equal — the matrix is symmetric.

□ □ 2

3. For C above andx = □−1□ compute Cx both ways. 1 □−1□

[Source: LA Lecture, week 3, problem 3]

Solution. Rows: 1·2+2·(−1)+(−1)·1 = −1; 0+3(−1)+2 = −1; so C**x** =. −1

rows. CTpoKH: 1-2+2-(-1)+(-1)-1 = -1; 0+0-(-1)+2 = -1; 3HAUYUT COX = [1]: [17 9; a | P14]

Columns: 2 − + =. Dimensions (2 × 3)(3 × 1) give output 2 × 1.

C = [0 3 2; -1 0 0; 0 -1 0]

4. With A, B from problem 1 compute AB and BA. Are they equal?

[5 1 2; -2 -1 1; 1 -2 4]

[Source: LA Lecture, week 3, problem 4]

Solution. AB =

[6 3; 8 7]

, BA =

[8 10; 0 0]

. Different; even when both orders exist, equality is not guaranteed.

[2 0 0; 0 1 -1; 0 1 1], D = [1 -1 0; 0 1 1; 0 0 1]

5. Let C = [1 2 −1; 0 3 2], D = [2 0; 1 −1; 3 2]. Predict the sizes of CD and DC, then compute both.

[Source: LA Lecture, week 3, problem 5]

Solution. CD: (2 × 3)(3 × 2) = 2 × 2; DC: (3 × 2)(2 × 3) = 3 × 3. CD = [2 + 2 − 3, 0 − 2 − 2; 0 + 3 + 6, 0 − 3 + 4] = [1 −4; 9 1]; DC = [2 4 −2; 1 −1 −3; 3 12 1].

6. Let S = (shift), H = (stretch),x =. Compute HSx and

[0 1 0; 1 1]

SHx; describe each transformation.

[2 4; 2 3]

[Source: LA Lecture, week 3, problem 6]

Solution. S**x** =

[1 1; 0 1]

, then HS**x** =

[2 4; 2 3]

; also H**x** =

[2 4; 2 3]

, then SH**x** =

[0 1 0; 1 1]

. In HS the shift stretch comes after; in SH the stretch happens first, then the shift is added on top. Same two operations, different order, different result: matrices don't commute.

[1; 2]

,**v** =

[2; 1]

: compute$\mathbf{u}^{\mathrm{T}}\,\mathbf{v}$ and$\mathbf{u}\mathbf{v}^{\mathrm{T}}$; check $(\mathbf{u}\mathbf{v}^{\mathrm{T}})^{\mathrm{T}} = \mathbf{v}\mathbf{u}^{\mathrm{T}}$.

7. Foru = 2 0 −1 3

[Source: LA Lecture, week 3, problem 7]

Solution. Inner: $\mathbf{u}^{\mathrm{T}}\,\mathbf{v} = 2 + 0 - 3 = -1$ (scalar). Outer: **uv^{T** =

[2 0 -1; 0 0 0; 3 0 -3]

(3 × 3). Transposing element-wise, $(\mathbf{u}\mathbf{v}^{\mathrm{T}})^{\mathrm{T}} = (\mathbf{v}\mathbf{u}^{\mathrm{T}})_{\mathrm{ij}}$, so $(\mathbf{u}\mathbf{v}^{\mathrm{T}})^{\mathrm{T}} = \mathbf{v}\mathbf{u}^{\mathrm{T}}$. The written matrices are 3 × 3 transposed to each other.

$$(AB)^{T} = B^{T}A^{T}$$

.

8. Using A, B from problem 1, compute $(\mathbf{AB})^{\mathbf{T}}$ and $\mathbf{A}^{\mathbf{T}}\,\mathbf{B}^{\mathbf{T}}$ independently; does $\mathbf{A}^{\mathbf{T}}\,\mathbf{B}^{\mathbf{T}}$ match?

[4] 4 −7 6

[Source: LA Lecture, week 3, problem 8]

Solution. AB =

begin{bmatrix} 3 & 12 -10 & -14 end{bmatrix}

, so $(\mathrm{AB})^{\mathrm{T}}$ =

begin{bmatrix} 3 & -10 12 & -14 end{bmatrix}

. First

begin{bmatrix} 6 & 3 -7 & 3 end{bmatrix} begin{bmatrix} 1 & 2 -1 & 0 end{bmatrix} = begin{bmatrix} 3 & 12 -10 & -14 end{bmatrix}

transposed: A^T =

$$\begin{bmatrix} -2 & 0 \\ 1 & 4 \end{bmatrix}$$

, B^T =

$$\begin{bmatrix} 4 & -7 \\ 6 & 3 \end{bmatrix}$$

; then $A^T B^T$ =

$$\begin{bmatrix} -2 & 0 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} 4 & -7 \\ 6 & 3 \end{bmatrix} = \begin{bmatrix} -8 & -14 \\ 28 & -5 \end{bmatrix}$$

. Computing: $(2 \cdot 1 + (-1)(-2)) = 4$, $(2 \cdot 3 + 0) = 6$

$$\begin{bmatrix} 4 & 6 \\ 28 & -5 \end{bmatrix}$$

$= (AB)^T$, which confirms (70). But $A^T B^T$ =

$$\begin{bmatrix} -8 & -14 \\ 28 & -5 \end{bmatrix}$$

is different: the correct identity changes the order of factors.

9. Let E = [3 1; 1 2]. Compute det E; find the areas of the parallelogram and triangle spanned by its columns; is orientation preserved?

[Source: LA Lecture, week 3, problem 9]

Solution. det E = 3·2 − 1·1 = 5. Parallelogram area = |5| = 5; triangle area = 5/2. Orientation is preserved: det E > 0.

10. Compute det F for F = [0 3 1] by cofactors of the first row; check by the first column.

$$\begin{bmatrix} 3 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 3 & 1 \end{bmatrix}$$

[Source: LA Lecture, week 3, problem 10]

Solution. First row: det F = 2 det −1 det +0 = 2(6)−1(−1) = 13. 0 2 1 2

$$\det F = 3 \det \begin{bmatrix} 0 & 2 \\ 1 & 0 \end{bmatrix} - 1 \det \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} + 0 \det \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 3(0) - 1(0) + 0(1) = 0. \quad (3.1)$$

First column: 2 det + 1 det = 12 + 1 = 13 (cofactor sign (3, 1) 0 2 3 1 is $(-1)^{3+1}$ = +). Any row or column gives the same value.

11. For a 3 × 3 matrix M, det M = 5 is known. Form N: swap rows 1,2; multiply row 3 by 2; add 3×(new row 1) to row 2. Find det N, det(2M), det(M^T M).

[Source: LA Lecture, week 3, problem 11]

Solution. Permutation: sign changes (−5); scaling row 3: another factor of 2; adding a row: no change. det N = $-2 \cdot 5 = -10$. Scaling all rows: det(2M) = $2^3 \cdot 5$ = 40. Product rule: det(M^T M) = det M^T det M = 25.

$$\det(M^T M) = \det M^T \det M = 25.$$

= [0],**v** = [2],**w** = [1]. Compute (**u** ×**v**) ·**w**, volumes

12. Letu 1 0 3 of the parallelepiped and tetrahedron; what changes if we swapu,v?

$$\begin{bmatrix} 2 \cdot 2 - 0 \cdot 1 & 4 \end{bmatrix}$$

[Source: LA Lecture, week 3, problem 12]

Solution. u × v = [1 · 1 − 2 · 0; 2 · 0 − 1 · 3; 1 · 3 − 2 · 0] = [1; -3; 3]; dot with w: 1 · 2 + (-3) · 1 + 3 · 4 = 2 - 3 + 12 = 11. Volume

$$2 \cdot 2 - 0 \cdot 1\ 4$$

of parallelepiped 11; of tetrahedron $\frac{11}{}$. Swapping u and v changes the sign to -11, but both volumes remain 11 and $\frac{11}{6}$ (absolute values).

$$0 \cdot 0 - 1 \cdot 2 = -2$$

.

13. Foru 1 1 k coplanar; expressw in terms ofu,v. □ □ □ □

$$0 \cdot 1 - 1 \cdot 2 - 2$$

⇐⇒ T = 0. Hereu ×v = □1 · 0 − 1 · 1□ = □−1□,

[Source: LA Lecture, Week 3, Problem 13]

Solution. Coplanarity

$$1 \cdot 2 - 0 \cdot 0\ 2$$

means T = −2 − 1 + 2k = 2k − 3 = 0 ⇒ k = $\frac{3}{2}$. Then**w** = α**u** + β**v**: the first two coordinates give α = 1, 2β = 1 ⇒ β = $\frac{1}{2}$; checking the third: 1 + 1_2 = 3_2 — consistent,**w** =**u** + $\frac{1}{2}$ **v**.

$$W = U + (5/7)V,\ [2\ 0; 0\ 1]D = [1\ 0; 0\ 1]\ (16)$$

14. Let D = [2 0; 0 3], X = [a b; c d]. Find all X with DX = XD; what if both diagonal elements of D are equal?

[Source: LA Lecture, Week 3, Problem 14]

Solution. DX = [2a 2b; 3c 3d], XD = [2a 3b; 2c 3d]. Equality requires b = c = 0, with a and d free: precisely diagonal matrices commute with this D. If D = λI, then DX = λX = XD for any X.

15. Letu = □0□,v = □2□,w = □1□ and L = □0 1 0 □. Without recalculating the edges,

$$[1\ 0\ 3; 0\ 0\ -1]$$

find the signed mixed product and volume after applying L to each edge; why is the volume factor universal?

[Source: LA Lecture, Week 3, Problem 15]

Solution. Let U = [**u v w**]; from problem 12, det U = 13. The transformed edges form LU, so det(LU) = det L det U = (−2)(13) = −26; new volume 26. The factor | det L| = 2 is independent of U, because det(LU) = det L det U for any U — the determinant product rule has geometric meaning: |det L| is the volume scaling factor of the transformation.

5 Week 4 — Matrix Rank and Inverse Matrix (Linear Algebra I)

Week 4 of Linear Algebra I covers matrix rank and matrix inversion via Gauss-Jordan elimination, culminating in Test 1 on weeks 1–3. There is no separate exercise sheet for week 4 in the repository, so this section provides (i) a complete catalog of formulas on the two topics, compiled from the assigned Strang material, and (ii) a graded set of practice problems in the style of the course's TA sheets, with full solutions. They use the same mathematical structures as elsewhere in the course, thus serving simultaneously as preparation for Test-1 on the matrix-algebra parts.

5.1 Formula catalog for week 4

6.1.1 Rank and elimination

rank A ≤ min(m, n), (106) $A \in R^{n\times n}$: rank A = n ⟺⟹ A**x** =**0** has only**x** =**0**, (107) A is invertible ⟺⟹ rank A = n ⟺⟹ n pivots ⟺⟹ det $A \neq 0$, (108) A is invertible ⟺⟹ A**x** =**b** has exactly one solution for any**b**. (109)

6.1.2 Inverse matrix (Strang, Introduction to Linear Algebra 4e, §2.5)

$AA^{-1} = I = A^{-1}A$; **x** = A^{-1}**b** solves A**x** =**b**, (110) A^{-1} exists ⟺⟹ elimination gives n pivots, (111)

$$[a\ b;\ c\ d]^{-1} = (1)/(ad - bc)\ [d\ -b;\ -c\ a]\ (ad - bc \neq 0),\ (112)$$

(reverse order), (113) □ 1 □ diag(d_1, . . . , d_n)$^{-1}$ = diag

$$diag(d_1, ..., d_n)^{-1} = diag(1/d_1, ..., 1/d_n)\ (d_i \neq 0),\ (114)$$

x≠**0**, A**x** =**0** ⟹ A not invertible, (115) BA = I, AC = I ⟹ B = B(AC) = (BA)C = C (uniqueness). (116) Interpretation (115): no matrix can map**0** back to**x**.

6.1.3 Gauss-Jordan elimination

$$elementary\ rows\ _{-1} \longrightarrow [\ I\ A]$$

[A I] (117) The same row operations turn A into I and I into A^{-1}. Row operations: swap two rows; multiply a row by a nonzero constant; add a multiple of one row to another. Each corresponds to an elementary matrix E with simple inverse (undo the operation), so $(E_2E_1A)^{-1} = A^{-1}E_1{}^{-1}E^{-1}$ comes out in reverse order. 2

5.2 Practice problems for week 4 (rank, inverse, Gauss-Jordan) — questions and full solutions

□ □ 1 2 1

1. Compute rank A for A = □2 4 5□ by elimination. 3 6 4

[Source: LA Practice Week 4, Problem 1]

Solution. Eliminate under the first pivot (1): $R_2 - 2R_1 \to (0, 0, 3)$; $R_3 - 3R_1 \to (0, 0, 1)$. The second column entries vanished, so the second pivot must come from column 3: pivot = 3. Eliminate under it: $R_3 - \frac{1}{3} R_2 \to (0, 0, 0)$. Two nonzero pivots: rank A = 2. Note the dependence $col_2 = 2col_1$, visible from the start — a rank deficiency signal. □2 □ 1

2. Determine if B = is invertible by three methods: pivots, 4 2 determinant, and nonzero null-vector.

[Source: LA Practice Week 4, problem 2]

Solution. (Pivots) $R_2 - 2R_1 \to (0, 0)$: only one pivot — degenerate. (Determinant) $ad - bc = 2 \cdot 2 - 1 \cdot 4 = 0$ — degenerate by (112). (Zero-vector) $B(1, -2)^T = (2 - 2, 4 - 4)^T = \mathbf{0}$ — non-zero solution to $B\mathbf{x} = \mathbf{0}$, so by (115) B is not invertible. All three tests agree. □ 2 □ 1 by the 2 × 2 formula and check $AA^{-1} = I$.

3. Find the inverse of A = 5 3 □ 3 □ −1

[Source: LA Practice Week 4, problem 3]

Solution. det A = 6 − 5 = 1, so by (112) A^{-1} = . Check: −5 2

$$A^{-1} = (1/7) \cdot [2 -3; 1 5]$$

=. □

$$A = [5 \cdot 3 + 3 \cdot (-5) \; 5 \cdot (-1) + 3 \cdot 2; 0 \; 1] = [0 \; 1; 0 \; 1]$$

4. Apply Gauss-Jordan to [A I], to invert A =. 3 7

[Source: LA Practice Week 4, problem 4]

Solution.

$$[1 \; 2; 3 \; 7] \to R_2 - 3R_1 \; [1 \; 2; 0 \; 1] \to R_1 - 2R_2 \; [1 \; 0; 0 \; 1], \; A^{-1} = [-7 \; 2; 3 \; -1]$$

So A^{-1} =; check: AA^{-1} = =. □

$$K = (1)/(-3 \cdot 2 - 1 \cdot (-1) + 0 \cdot (-1)) \; begin\{bmatrix\} 2 \; \& \; -1 \; \& \; 0 \; -1 \; \& \; 2 \; \& \; -1 \; 0 \; \& \; -1 \; \& \; 2 \; end\{bmatrix\} = 1$$

5. Invert by Gauss-Jordan and check that 0 −1 2 the product of pivots equals det K.

[Source: LA Practice Week 4, problem 5 (Strang's favorite example)]

Solution. Work on [K I]: eliminate with first pivot 2 ($R_2 + \frac{1}{2} R_1$), then with second pivot $\frac{3}{2}$ ($R_3 + \frac{2}{3} R_2$), arriving at upper triangular form with pivots 2, $\frac{3}{2}$, $\frac{4}{3}$. Then clean upward and divide each row by its pivot; the right block becomes □ □ 3 2 1

$$K^{-1} = (1/4) \; [2 \; -1 \; 2; 0 \; 1 \; 0; -1 \; 0 \; 1]$$

Product of pivots is $2 \cdot \frac{3}{2} \cdot 4 = 4 = \det K$, and indeed K^{-1} divides the integer matrix by det K — the pattern noted after (112). (K is symmetric and tridiagonal, but K^{-1} is dense.)

$$A = [1 \; 2; 1 \; 1]$$

using A^{-1} from problem 4.

6. Solve Ax =b for A =,b = 3 7 1

$$[7 \; -2; 1 \; 1] \; [5; -2]$$

[Source: LA Practice Week 4, problem 6]

Solution. x = A^{-1}b =

$$begin\{bmatrix\} -3 \; \& \; 1 \; 1 \; \& \; 1 \; end\{bmatrix\}$$

. Check: 1 · 5 + 2(−2) = 1 and 3 · 5 + 7(−2) = 1. □

$$A^{-1} = (1)/(7) \; begin\{bmatrix\} 1 \; \& \; -2 \; 3 \; \& \; 1 \; end\{bmatrix\}$$

; compute $(AB)^{-1}$ without first

7. Given A = and B = 0 2 1 1 multiplying, then check.

[Source: LA Practice Week 4, problem 7]

Solution. By (113), $(AB)^{-1} = B^{-1}A^{-1}$. Here det A = 2 and det B = 2, so

$$A_{-1} = [1\ 0; 0\ 1],\ B_{-1} = [(-1\ 2\ 1)/2\ (2\ 1)/(-2); (-3\ 1\ 4)/4\ (2\ 1)/(-2)],\ det(AB) = 4,\ (AB)^{-1} = 1/4\ [2\ -1; -3\ 1]$$

, det(AB) = 4, so directly $(AB)^{-1}$ = 1/4 [2 −1; −3 1]. Check: AB = [1 0; 0 1] * [(−1 2 1)/2 (2 1)/(−2); (−3 1 4)/4 (2 1)/(−2)] =

$$det(AB) = 4,\ find\ the\ inverse\ matrix\ (AB)^{-1} = (1)/(4)\ begin\{bmatrix\}\ 2\ \&\ -3\ -2\ \&\ 4\ end\{bmatrix\}$$

[1 0; 0 1] *[(−1 2 1)/2 (2 1)/(−2); (−3 1 4)/4 (2 1)/(−2)] = [1/2 1/2; 3/4 1/4] — the same matrix. And $(AB)^{-1}(AB)$ = [1/2 1/2; 3/4 1/4]* [1 0; 0 1] = [1/2 1/2; 3/4 1/4] * [(−1 2 1)/2 (2 1)/(−2); (−3 1 4)/4 (2 1)/(−2)] = [1 0; 0 1]. Reverse order law confirmed.

8. (Conceptual, Test-1 style.) True or false, with justification: (a) If A is invertible, then A^T is invertible. (b) If Ax = λx with x≠0 and λ = 0, then A is not invertible. (c) The product of invertible 3 × 3 matrices can be singular. (d) rank(A) ≤ 2 for any 2 × 3 matrix.

[Source: LA Practice Week 4, problem 8]

Solution. (a) True: A has n pivots ⟺ A^T has n pivots (row rank = column rank); formally $(A^T)^{-1} = (A^{-1})^T$. (b) True: non-zero **x** with A**x** = **0** = 0 · **x** = **0** violates (115). (c) False: by (113) the product of invertibles is invertible. (d) True: rank does not exceed the number of rows (each pivot uses one row), so rank ≤ 2 < 3.

9. (Applied, Test-1 style.) For what value(s) of c is the matrix A = 1 4 2 singular? For other values solve Ax = by the inverse 3 c 10 formula. c −2

[Source: LA Practice Week 4, problem 9]

Solution. det A = c − 6; singular exactly when c = 6. Otherwise A^{-1} = 1

$$det\ A = (c - 6)/(4c - 20)$$

and**x** = A^{-1}**b** = 1, for example when c = 7:**x** = $(8, -2)^T$.

$$x = A^{-1}b = (begin\{array\}\{c\}8-2end\{array\}).$$

10. (Advanced difficulty.) Show that if A is invertible, then rank(AB) = rank(B) for any compatible B.

[Source: LA Practice Week 4, Problem 10]

Solution. Multiplication by an invertible matrix does not change the number of pivots: standard reasoning: rank(AB) ≤ rank B (rows of AB are combinations of rows of B), and applying the same to B = A^{-1}(AB), we get rank B ≤ rank(AB); both inequalities force equality. Intuition: an invertible factor only stretches/rotates/shifts spaces, but does not flatten them, so rank cannot be lost or gained.

6 Textbook supplements — additional required reading exercises with full solutions

6.1 Strang, Introduction to Linear Algebra 4e — selected problems solved

1. (§1.1) Describe all cu for u = (2, 1) and all combinations cu + dv for u = (2, 1), v = (1, 2); when do the combinations cu + dv fill the plane?

[Source: Strang Intro 4e, Concept Set §1.1]

Solution. c**u** fills the line through (2, 1): every vector on it is a multiple. c**u** + d**v** fill the plane through the origin, unless **u** and **v** are collinear; here det = 3 ≠ 0, so the two directions are independent and the combinations fill all R^2. This very criterion — nonzero determinant — returns in week 4 as invertibility (108).

2. (§1.2) Compute the dot products: u ·v for u = (1, 2, 1), v = (3, 0, 2); find a vector w perpendicular to both.

[Source: Strang Intro 4e, Problem Set §1.2]

Solution. u ·v **= 3 + 0 + 2 = 5. A perpendicular vector is the cross product:** w **=** u **×** v** =

$$[1; 2; 1] \times [3; 0; 2] = [(2\cdot 2 - 1\cdot 0); (1\cdot 3 - 1\cdot 2); (1\cdot 0 - 2\cdot 3)] = [4; 1; -6]$$

; check: **w** ·**u** = 4 + 2 − 6 = 0 and **w** ·**v** = 12 + 0 − 12 = 0. □

3. (§1.2) When is u ·v = v ·u negative and what does that say about the angle? When is it zero?

[Source: Strang Intro 4e, Concept §1.2]

Solution. Negative exactly when cos θ < 0, i.e., when the angle is greater than 90° (obtuse); zero when θ = 90° — perpendicularity. This is the classification used in TA problems of week 2. □2 □ 1 , **b** = $(3, 4)^T$ through A^{-1}.

4. (§2.5) Solve Ax = b with A = 1 2

$$[2 -1 2; 3 -4 2; 2/3]$$

[Source: Strang Intro 4e, Problem Set §2.5]

Solution. det A = 3: A^{-1} = ^{1}; **x** = A^{-1}**b** = ^{1} = ^{1} =.

$$detA = 3; A = [1 -1 2; 3 -3 8; 3 5 5/3]; x = A^{-1}b = [1/4; 4; 2] = [14/4; 4; 2] = [7/2; 4; 2] \ (16)$$

Check: 2 · $^2_3$ + 5_3 = 3, 2_3 + 2 · 5_3 = 4. □

5. (§2.5) Why does a nonzero solution to Ax = 0 rule out invertibility? Give the two-sentence reasoning that the notes call "no matrix can map 0 back to x."

[Source: Strang Intro 4e, Note 4 to §2.5]

Solution. If A**x** = **0** with **x** ≠ **0** and A had an inverse, then **x** = A^{-1}A**x** = A^{-1}**0** = **0** — contradiction. The zero vector has no "preimage" only when the mapping is one-to-one; the zero vector proves that it is not.

6. (§2.5) If A and C are invertible 3 × 3 matrices, is A + C invertible? What about AC? What about $A^{-1}C$?

[Source: Strang Intro 4e, Problem Set §2.5]

Solution. A + C: not necessarily (for example, C = −A gives 0); sums of invertible matrices can be singular. AC: yes, with $(AC)^{-1} = C^{-1}A^{-1}$ (113). $A^{-1}C$: yes, product of two invertible matrices.

6.2 Strang, Linear Algebra and its Applications 4e, §1.6 (Inverses and Transposes) — key results and worked example

Assigned reading §1.6 adds three items to the catalog: Gauss-Jordan elimination for A = □ □ 2 1 1

$$det\ A^{-1} = (1)/(det\ A)$$

and the identities −2 7 2 of transposition. □ □ 2 1 1

1. Perform Gauss-Jordan elimination on A = □ 4 −6 0□ and find A^{-1}. −2 7 2

[Source: Strang LAiA 4e, §1.6, example 1]

Solution. We complete I and eliminate: pivots 2, −8, 1 (after $R_2 - 2R_1$, $R_3 + R_1$, then $R_3 + \frac{3}{4} R_2$ over the reduced rows). Continuing upward, clearing above pivots and dividing rows by 2, −8, 1, we get □ □ □ □

$$A^{-1} = (1/8)\ [12\ -5\ -6;\ 8\ -3\ -2;\ -1\ 1\ 1]$$

The determinant equals the product of pivots 2(−8)(1) = −16, and indeed A^{-1} shows division by det A (accounting for cofactor placement — equivalent to $A^{-1} = \frac{\text{adj}A}{\det A}$).

2. State the determinant-transposition identities used in §1.6: det A^T, det(AB), det A^{-1}.

[Source: Strang LAiA 4e, §1.6]

Solution. $\det A^T = \det A$; $\det(AB) = \det A \det B$ (table 1); and, setting $B = A^{-1}$, $\det(A) \det(A^{-1}) = \det I = 1$, so $1 \det A^{-1} = \det A$, which precisely means "invertible ⟺ det $A \neq 0$", connecting the catalog of determinants from week 3 with the catalog of inverses from week 4.

3. Prove $(A^{-1})^T$ $(A^T)^{-1}$ why $A^T A$ is symmetric with = and explain, nonnegative diagonal.

[Source: Strang LAiA 4e, §1.6]

Solution. Transposing $AA^{-1} = I$, we get $(A^{-1})^T A^T = I$, which means

$$(A^{-1})^T = (A^T)^{-1}\ (16)$$

precisely: Symmetry: $= A^T A$; diagonal elements equal P

$$(A^T)^{-1} = (A^{-1})^T;\ \textit{the matrix } A^T A \textit{ is symmetric, } (A^T A)^T = A^T A,\ \textit{its diagonal elements are nonnegative:}$$
$$a_{ii} = \Sigma_{k=1}^{n} a_{ki}^2 \geq 0 \text{ — sums of squares — a fact. (17)}$$

behind CC^T in the lecture problems of week 3.